

A non-rigorous, non-mathematical, but hopefully
intuitive overture on
Brownian Motion, Stochastic Integral and Ito's
Formula

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Outline

Brownian Motion / Wiener Process

Information, Martingality, Markovianity

Stochastic Integral

Itô Formula

Diffusion Processes and Stochastic Differential Equations

Multidimensional case

Multivariate SDEs

Brownian Motion / Wiener Process

The Brownian Motion - Wiener Process

A *stochastic process* is a family of random variables, $X = X(t)$, $t \in \mathcal{T}$ that describes the evolution of an uncertain quantity in time.

Discrete-time stochastic processes take $\mathcal{T} = \{t_0, t_1, \dots, t_N \dots\}$ but we will work with *continuous-time stochastic processes* taking $\mathcal{T} = [0, T]$ or $\mathcal{T} = [0, \infty)$.

Definition

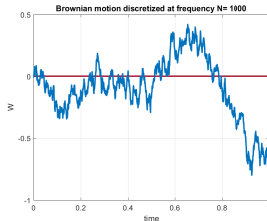
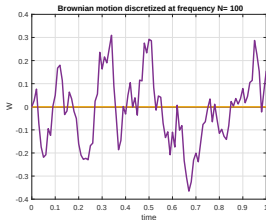
A stochastic process W is called a *standard Brownian motion* (**Wiener process**) if:

- ▶ $W(0) = 0$ a.s. (this means $\mathbb{P}(W(0) = 0) = 1$);
- ▶ normally distributed increments: for all $t > s \geq 0$:
 $W(t) - W(s) \sim \mathcal{N}(0, t - s)$;
- ▶ independent increments: for all $r < s \leq t < u$ we have that
 $W(s) - W(r) \perp W(u) - W(t)$;
- ▶ (W has a.s. continuous paths).

Remarks

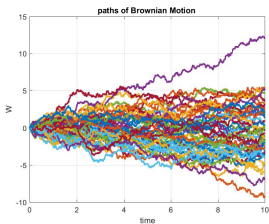
- ▶ Clearly $E[W(t)] = 0$ and $\text{Var}[W(t)] = t$;
- ▶ To simulate the path of W at n equidistant points $0 = t_0 < t_1 < t_2 < \dots < t_n = t$, $\Delta t = t_{j+1} - t_j$ we may proceed as follows
 - Generate the increments, that is $\phi_{j=1,\dots,n}$, i.i.d $\sim \mathcal{N}(0, 1)$
 - Start at $W(0) = 0$,
 - $W(t_{j+1}) = W(t_j) + \sqrt{\Delta t} \phi_j$

- Path simulations can be found in the script "Companion01BrownianMotion"



Although continuous, trajectories are quite "jagged".

- A bunch of 50 different paths simulated over 10 years looks as follows



p-Variation

- ▶ The p-variation gives a measure of "jaggedness" of a process. Given a process X , the p-variation over $[0, t]$ is defined as

$$|X|^p(t) = \lim_{\Pi \rightarrow 0} \sum_{j=1}^N |X(t_j) - X(t_{j-1})|^p$$

where $\Pi = \sup_{j=1, \dots, N} (t_j - t_{j-1})$, denotes the mesh-size of the partition of $0 = t_0 < t_1 < \dots < t_N = t$.

- ▶ We refer to the 1-variation as "variation" and to the 2-variation as the "quadratic variation", which is often denoted by $[X](t)$.
- ▶ The paths of the Wiener process W are a.s. of infinite variation on any $[0, t]$ and in particular, nowhere differentiable.
- ▶ On the other hand, W has finite quadratic variation, and it holds that

$$|W|^2(t) = [W](t) = t$$

See again Companion01 for illustrations.

Realized Variance

- ▶ The quadratic variation of a process X is a fundamental quantity not only in stochastic analysis (which we will not fully appreciate here) but also in finance.
- ▶ $S(t)$ denote the price process of an asset. Consider a regular grid $0 = t_0 < t_1 < \dots < t_N = T$ on the time interval $[0, T]$. The associated *realized variance* is defined as

$$RV^N(T) = \sum_{j=1}^N (\log S(t_j) - \log S(t_{j-1}))^2,$$

that is the sum of the squared log-returns realized over the given time grid.

- ▶ A number of contracts are written on the realized variance:
 - ▶ Variance swaps: at maturity T a long position pays off $RV^N(T)/T - K$ where K is a fixed "variance-delivery" quantity.
 - ▶ Variance options: at maturity T a long call pays off $\left(\frac{RV^N(T)}{T} - K\right)^+$ where K is a fixed "variance-strike" quantity.
 - ▶ Volatility swaps, VIX options etc.

Realized Variance-2

- ▶ Typical contracts are based on daily frequency $\Delta t = 0.004$, which is a very small time interval.
- ▶ Common practice is to approximate realized variance with its - mathematically more tractable - continuous time limit, that is the quadratic variation of the log-price

$$RV^N(T) \approx [\log S](T)$$

- ▶ However, other sampling frequencies are possible, and the sampling dates are specified in the contract.

Examples/exercises

Assume the price of a non dividend paying asset is described by

$$S(t) = se^{\sigma W(t)}$$

where $W(t)$ is a standard Brownian motion and σ is a positive parameter. Today's spot price is $S(0) = s = 100$ and $\sigma = 0.2$.

1. Compute the expected (simple) return over a period of 3 months
 $T = 0.25$.

- 2 Consider an ATM European call option expiring in 3 months $T = 0.25$. Compute the probability of exercise and the expected payoff.

- 3 Consider a realized variance swap contract, expiring in 3 months $T = 0.25$, with monthly sampling dates $T_1 = 0.083$, $T_2 = 0.167$, $T_3 = T = 0.25$ and delivery $K = 0.04$. Compute the expected payoff from a long position.
- 4 Consider a realized variance call option with the same expiry and delivery as the swap above. Compute probability of exercise and the expected payoff.

- ▶ The answer to all the posed problems, can be solved using the definition of Brownian motion, elementary probability and few integral computations.
- ▶ The underlying model is simple and so are the involved payoffs.
- ▶ However, this is not always the case and one needs to resort to numerical methods to compute the required expectations.
- ▶ Besides, it is always very useful to have a safe-and-simple back-up method, to verify the validity of the mathematical formulae which you have derived.

Monte Carlo basic example

To estimate $E[g(X)]$ you may proceed as follows

- For $i = 1, \dots, n$, generate $X_1, \dots, X_n$ outcomes from X and compute the corresponding payoff $g(X_1), \dots, g(X_n)$;
- Estimate $E[g(X)]$ as the arithmetic average of the corresponding payoffs, i.e.

$$E[g(X)] \approx E_n = \frac{1}{n} \sum_{i=1}^n g(X_i). \quad (1)$$

- The standard error is given by

$$SE_n = \frac{1}{\sqrt{n}} \sqrt{\frac{1}{n-1} \sum_{i=1}^n (g(X_i) - E_n)^2} = \frac{SD_n}{\sqrt{n}}$$

where SD_n is the standard deviation of the sample.

- All very simple but the convergence is slow.

Path-dependent example

- Consider again Ex 1 & 2: We need to compute $E[g(W(T))]$ where g is given by

$$g(x) = e^{\sigma x}, \quad g(x) = (se^{\sigma x} - K)^+, \quad g(x) = I_{(se^{\sigma x} - K > 0)}$$

The MC solution is straightforward. You may find it in function "MCToyProbabilityExercise", "MCToyCall".

- ▶ How about an Asian contract which at maturity T pays out

$$\text{Payoff}(T) = (A(T) - K)^+ \quad \text{where} \quad A(T) = \frac{1}{N} \sum_{j=1}^N S(t_j)$$

with sampling dates $0 < t_1 < t_2 < \dots < t_n = T$ and strike K specified in the contract.

- ▶ To generate one outcome $A_i(T)$, we need to generate values of $W(t)$ at sampling points t_j , $j = 1, 2, \dots$, e.g. in accordance to slide 5
- ▶ A (possible) function for computing the expected payoff is given in "MCToyAsianCall".

► How about Ex 3 & 4?

► How about a continuously sampled Asian paying out

$$\text{Payoff}(T) = (A(T) - K)^+ \quad \text{where} \quad A(T) = \frac{1}{T} \int_0^T S(t) ds$$

Information, Martingality, Markovianity

Information and Filtrations.

- ▶ Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. That is
 1. Ω is the universe of possible outcomes.
 2. $\mathcal{F}$ represents the set of measurable events – where an event A is a collection of the universe outcomes. That is, $A \subseteq \Omega$, and $\mathcal{F} \subseteq 2^\Omega$ with some added properties which makes it a σ -algebra.
 3. $\mathbb{P}$ is the "true" or physical or statistical probability measure.

- ▶ In the probability space we also consider a **filtration** $(\mathcal{F}_t)_{t \geq 0}$ which models the evolution of information through time.
- ▶ $\mathcal{F}_t \subseteq \mathcal{F}$ which contains all the events which are known by time t .
- ▶ At time 0, we only have the trivial information $\mathcal{F}_0 = \{\emptyset, \Omega\}$. We can only say that we know nothing ($\emptyset$) and that anything can happen (Ω)
- ▶ As time passes, we gain information and the amount of events which we can say if they have occurred or not increases.
- ▶ A bit more rigorously we can define $(\mathcal{F}_t)_{t \geq 0}$ as a family of σ -algebras s.t.
 - ▶ $\mathcal{F}_t \subseteq \mathcal{F}$ for all $t \geq 0$.
 - ▶ $\mathcal{F}_t \subseteq \mathcal{F}_{t'}$ if $t \leq t'$.

In this course we will (almost always) identify $(\mathcal{F}_t)_{t \geq 0}$ with the filtration generated by a stochastic processes (often a Brownian motion).

Definition 4.2

- ▶ Let X be a process. The symbol $\mathcal{F}_t^X$ denotes the information generated by X over the interval $[0, t]$, or alternatively "what has happened to X over the interval $[0, t]$ ".
- ▶ An event $A \in \mathcal{F}_t^X$ if, based upon the path $\{X(s); 0 \leq s \leq t\}$ one can decide if A has occurred or not. In such a case we say that " A is $\mathcal{F}_t^X$ - measurable".

Definition 4.2 - cont.

- ▶ If the value of a r.v. Z is fully determined by the trajectory $\{X(s); 0 \leq s \leq t\}$, we say that Z is $\mathcal{F}_t^X$ measurable and write $Z \in \mathcal{F}_t^X$.

- ▶ Let Y be a stochastic process. If $Y(t) \in \mathcal{F}_t^X$, for all $t \geq 0$, then we say that Y is adapted to the filtration $(\mathcal{F}_t^X)_{t \geq 0}$

Examples

Let W be a Brownian Motion and let $S(t) = S(0)e^{\sigma W(t)}$ the price of an asset.

- ▶ The event "a European call is exercised", i.e. $A = \{S(T) > K\}$. Then $A \in \mathcal{F}_T^W$.
- ▶ The process $S(t)$ is adapted to $(\mathcal{F}_t^W)_{t \geq 0}$.
- ▶ In fact, $W(t)$ is adapted to $(\mathcal{F}_t^S)_{t \geq 0}$.
- ▶ The integrated process $I(t) = \int_0^t W_s ds$ is adapted to $(\mathcal{F}_t^W)_{t \geq 0}$.

- ▶ Is $W(t)$ adapted to $(\mathcal{F}_t^I)_{t \geq 0}$?

- ▶ Björk, p. 43 for more examples.

Conditional expectation: Operations Sheet

Consider a given filtration $(\mathcal{F}_t)_{t \geq 0}$. You may think of $(\mathcal{F}_t^W)_{t \geq 0}$.

1. For a r.v. Y , we let the symbol

$$E[Y|\mathcal{F}_t]$$

denote the "*expected value of Y given the information up to time t* ".
In short, the conditional expectation of Y given $\mathcal{F}_t$.

2. $E[Y|\mathcal{F}_t]$ is a stochastic variable, and $E[Y|\mathcal{F}_t] \in \mathcal{F}_t$.
3. If $Y \in \mathcal{F}_t$, then $E[Y|\mathcal{F}_t] = Y$ and if $Y \perp \mathcal{F}_t$ then $E[Y|\mathcal{F}_t] = E[Y]$.

4 Let Z be a r.v. Then

$$E[Z + Y|\mathcal{F}_t] = E[Z|\mathcal{F}_t] + E[Y|\mathcal{F}_t]$$

and if $Z \in \mathcal{F}_t$, then

$$E[ZY|\mathcal{F}_t] = ZE[Y|\mathcal{F}_t]$$

5 Law of iterated expectations: If $s < t$, then

$$E[E[Y|\mathcal{F}_t]|\mathcal{F}_s] = E[Y|\mathcal{F}_s]$$

Examples

Let W be a Brownian Motion, and let $s \leq t$ and let $(\mathcal{F}_t^W)_{t \geq 0}$ be the filtration generated by W .

- ▶ $E[W(t)|\mathcal{F}_s^W] = W(s)$
- ▶ $E[W^2(t)|\mathcal{F}_s^W] = W^2(s) + (t - s)$

► $E \left[e^{\alpha W(t)} | \mathcal{F}_s^W \right] = e^{\frac{\alpha^2}{2}(t-s) + \alpha W(s)}$

► Let $I(t)$ denote the integrated Brownian motion process, i.e.

$$I(t) = \int_0^t W(s) ds$$

then

$$E \left[I(t) | \mathcal{F}_s^W \right] = I(s) + W(s)(t - s)$$

Martingales.

Definition

A stochastic process X is called an $\mathcal{F}_t$ -**martingale** if the following conditions hold.

- ▶ X is adapted to $\mathcal{F}_t$;
- ▶ $E[|X(t)|] < \infty$ for all $t > 0$;
- ▶ For all s and t with $s \leq t$ the following relation holds $E[X(t)|\mathcal{F}_s] = X(s)$.

Definition

A process X satisfying, for all s and t with $s \leq t$, the inequality

$$E[X(t)|\mathcal{F}_s] \leq X(s),$$

is called an $\mathcal{F}_t$ -**supermartingale**, and a process X satisfying

$$E[X(t)|\mathcal{F}_s] \geq X(s),$$

is called an $\mathcal{F}_t$ -**submartingale**.

Examples

Let W be a Brownian Motion. With the results of Slide 27 at hand, it should be fairly easy to see that

- ▶ The Brownian motion W is a martingale.
- ▶ $X(t) = W^2(t) - t$ is a martingale

► $X(t) = e^{-\frac{\alpha^2}{2}t + \alpha W(t)}$ is a martingale

Markov Processes – the slide of shame

Consider a process X and a filtration $(\mathcal{F}_t)_{t \geq 0}$.

- ▶ **Sloppy definition!** We say that a process X is a *Markov* process (is Markovian - has the Markov property) if

"The future is independent of the past given the present."

that is for any $0 \leq s < t$, it holds that

$$X_t \perp \mathcal{F}_s | X_s$$

- ▶ Trying to be slightly more concrete

$$X(t) | \mathcal{F}_s \text{ has the same distribution as } X(t) | X(s).$$

That is: the past history up to s contains no information about the future of the process that cannot be extracted from the current state $X(s)$.

- And trying to be a tiny bit more precise

$$\mathbb{E}[f(X(t)) | \mathcal{F}_s] = \mathbb{E}[f(X(t)) | X(s)], \quad \forall \text{ function } f \text{ and } t \geq s$$

- In particular

$$E[f(X(t)) | \mathcal{F}_s] = F(s, t, X(s))$$

Examples

Let W be a Brownian Motion, and consider the filtration $(\mathcal{F}_t^W)_{t \geq 0}$.

- ▶ W is Markovian.
- ▶ $S(t) = S(0)e^{W(t)}$ is Markovian.
- ▶ The integrated Brownian motion $I(t) = \int_0^t W(s)ds$ is **NOT** Markovian
- ▶ However, the pair process (W, I) is Markovian

Stochastic Integral

Stochastic Integral

Consider a Wiener process W and a stochastic process g adapted to $\mathcal{F}^W$, and such that $g \in \mathcal{L}^2[a, b]$, i.e.

$$\int_a^b E[g^2(s)]ds < \infty. \quad (2)$$

Then the stochastic integral $\int_a^b g(s)dW(s)$, is "defined" as the L_2 -limit of the Riemann sums

$$\int_a^b g(s)dW(s) = \lim_{\Delta_n \rightarrow 0} \sum_{k=0}^{n-1} g(t_k)(W(t_{k+1}) - W(t_k)) \quad (3)$$

where $\Delta_n = \sup_{k=0, \dots, n-1} (t_{k+1} - t_k)$, denotes the mesh-size of the partition of $a = t_0 < t_1 < \dots < t_n = b$.

Notice that in (3), the increments of W are taken **forward**, i.e. the stochastic integral is defined by **non-anticipative** sums.

- The "definition" of the stochastic integral given above is a "proposition" rather than a definition. For a more precise definition, see Björk.

Natural properties

- If $g(t) = g$ with $g \in \mathcal{F}_a^W$, then

$$\int_a^b g dW(s) = g \int_a^b dW(s) = g(W(b) - W(a))$$

- If $a < c < b$ then

$$\int_a^b g(s) dW(s) = \int_a^c g(s) dW(s) + \int_c^b g(s) dW(s)$$

► Let $g \in \mathcal{L}^2[0, t]$ for all $t \geq 0$ ($g \in \mathcal{L}^2$ for short).

► Then we can construct the stochastic integral process

$$X(t) = \int_0^t g(s) dW(s) \quad (4)$$

► $X(t)$ is adapted to $(\mathcal{F}_t^W)_{0 \leq t \leq T}$.

► To use a less cumbersome notation, we can re-write the stochastic integral process $X(t)$ in (4) in **stochastic differential** form, i.e.

$$dX(t) = g(t) dW(t) \quad X(0) = 0 \quad (5)$$

Examples

- Consider the process

$$X(t) = \int_0^t u \, dW(u) \quad \text{or} \quad dX(t) = t dW(t), \quad X(0) = 0. \quad (6)$$

Then

- $X(0) = 0$
- $X(t) - X(s) \sim \mathcal{N}(0, \int_s^t u^2 du)$ for all $t > s \geq 0$
- $X(s) - X(r) \perp X(u) - X(t)$ for all $r < s \leq t < u$;

- ▶ Consider the process

$$X(t) = \int_0^t W(u) dW(u) \quad \text{or} \quad dX(t) = W(t)dW(t) \quad (7)$$

- ▶ In this case X does not have independent increments. Also, we cannot say right away what is the distribution of $X(t)$.
- ▶ However, we could once again resort to a MC numerical scheme to compute e.g. $P(X(t) > K)$ or more general $E[F(X(t))]$
- ▶ If we know how to generate paths of the integrand process, the differential form in (7) tells us how to generate an **approximate** path of $X(t)$.

To simulate approximate paths of

$$X(t) = \int_0^t W(u) dW(u) \quad dX(t) = W(t)dW(t), \quad X(0) = 0,$$

- Choose a fine partition $0 = t_0 < t_1 < \dots < t_n = t$
- Start at 0, i.e. set $X(0) = 0$, and set $W(0) = 0$
- Recursively
 - generate increment of Brownian motion $\Delta W \sim \mathcal{N}(0, \Delta t)$
 - Set

$$\begin{aligned} X(t_{i+1}) &= X(t_i) + W(t_i) \cdot \Delta W \\ W(t_{i+1}) &= W(t_i) + \Delta W. \end{aligned}$$

Examples - Companion 02

Let $W(t)$ denote a BM.

- ▶ Assume that the price of an asset is given by $S(t) = S(0)e^{X(t)}$, where $X(t) = \int_0^t u \, dW(u)$.
- ▶ Then the log-returns over a small time interval $[t, t + \Delta t]$ are given by $\log(S(t + \Delta t)) - \log(S(t)) = X(t + \Delta t) - X(t) \approx t \cdot \Delta W(t) \sim \mathcal{N}(0, t^2 \Delta t)$.
- ▶ Although time inhomogeneous, they are independent of each other. See Companion 02 for simulation of log-price paths and log-returns

- ▶ Assume that the price of an asset is given by $S(t) = S(0)e^{X(t)}$, where $X(t) = \int_0^t W(u) \, dW(u)$.
- ▶ Then the log-returns over a small time interval $[t, t + \Delta t]$ are given by $\log(S(t + \Delta t)) - \log(S(t)) = X(t + \Delta t) - X(t) \approx W(t) \cdot \Delta W(t)$
- ▶ In this case, log-returns are inhomogeneous and dependent of each other. See Companion 02 for simulation of log-price paths and log-returns

Properties of the stochastic integral

Proposition (4.4)

Let g be an $\mathcal{L}^2[a, b]$ process. Then the following relations hold

- ▶ $E \left[\int_a^b g(s) dW(s) \right] = 0;$
- ▶ $E \left[\left(\int_a^b g(s) dW(s) \right)^2 \right] = E \left[\int_a^b g^2(s) ds \right], \text{ Ito Isometry};$

Remark 4.3.2 It is possible to define the stochastic integral for g satisfying only

$$P \left(\int_a^b g^2(s) ds < \infty \right) = 1.$$

However, under this weaker requirement there is no guarantee that Proposition 4.4 holds.

Sketch of proof

Proposition (4.7)

For any process $g \in \mathcal{L}^2[s, t]$ the following holds

$$E \left[\int_s^t g(u) dW(u) \middle| \mathcal{F}_s^W \right] = 0.$$

In particular, for any process $g \in \mathcal{L}^2$, the process X defined by

$$X(t) = \int_0^t g(s) dW(s),$$

is an $\mathcal{F}_t^W$ -martingale.

In other words, assuming the integrability condition, **every stochastic integral is a martingale.**

Sketch of proof

Itô Formula

Ito Process

Definition Let W be a Wiener process. An **Ito process** is a stochastic process of the form

$$X(t) = a + \int_0^t \mu(s)ds + \int_0^t \sigma(s)dW(s), \quad (8)$$

where $a \in \mathbb{R}$ and μ, σ are adapted and regular enough processes.

In differential form we write

$$dX(t) = \mu(t)dt + \sigma(t)dW(t) \quad X(0) = a,$$

highlighting that, over a small time-interval $[t, t + \Delta t]$

$$\Delta X(t) | \mathcal{F}_t \sim \mu(t)\Delta t + \mathcal{N}(0, \sigma^2(t)\Delta t).$$

So $\mu(t)$ represents the local expected rate of change, while $\sigma^2(t)$ represents the local rate of variance.

Examples

- ▶ Generalized BM

$$dX(t) = \mu dt + \sigma dW(t), \quad X(0) = x_0$$

- ▶ Time-inhomogeneous versions of the generalized BM

$$dX(t) = \mu(t)dt + f(t)dW(t), \quad X(0) = x_0$$

with $\mu(t)$ and $f(t)$ deterministic functions.

- ▶ More interestingly

$$dX(t) = \mu dt + \sigma W(t)dW(t), \quad X(0) = x_0$$

Problem Statement

- ▶ Let X be an Itô process

$$dX = \mu dt + \sigma dW$$

- ▶ Let f be some function of t and X . What is the stochastic differential for $f(t, X)$?
- ▶ In other words: given that we know dX , and obviously dt , what is $df(t, X)$?
- ▶ Before moving on, we still need a couple of "mathematical" tools.

Co-Variation

- **Definition** Let X and Y be two processes. The **co-variation** of X and Y up to time t is defined as

$$[X, Y](t) = \lim_{|\Pi| \rightarrow 0} \sum_{j=0}^{n-1} \left(X(t_{j+1}) - X(t_j) \right) \left(Y(t_{j+1}) - Y(t_j) \right),$$

where $|\Pi| = \sup_{j=0, \dots, n-1} (t_{j+1} - t_j)$.

- In particular the co-variation of a process X with itself is what we have defined as the quadratic variation of X , i.e.,

$$[X, X](t) = [X](t)$$

Let W be a Wiener Process. Then

$$[W, W](t) = [W](t) = t, \quad [W, t](t) = 0, \quad [t, t](t) = 0,$$

Informally, we may write the results above as

$$dWdW = dt, \quad dWdt = dtdW = 0, \quad dtdt = 0.$$

Heuristic "proof"

Theorem

The quadratic variation of the Itô process (8) is

$$[X](t) = \int_0^t \sigma^2(s) ds.$$

We drop the explicit time dependence in the notation, and write informally

$$dXdX = \sigma^2 dt.$$

This is justified by noticing that

$$\begin{aligned} dXdX &= (\mu dt + \sigma dW)^2 \\ &= \mu^2 \underbrace{dt dt}_{=0} + 2\mu\sigma \underbrace{dt dW}_{=0} + \sigma^2 \underbrace{dW dW}_{=dt} \\ &= \sigma^2 dt. \end{aligned}$$

Sum-up of "operational rules"

Let W be a standard Brownian motion, and let X be an Itô process. In short-hand notation

$$dX = \mu dt + \sigma dW$$

where μ and σ denote "regular enough" and adapted process.

Then

$$dWdW = dt, \quad dWdt = 0, \quad dt dt = 0$$

and

$$dXdX = \sigma^2 dt, \quad dXdt = 0, \quad (dt dt = 0)$$

Problem Statement

- ▶ Let X be an Itô process

$$dX = \mu dt + \sigma dW$$

- ▶ Let f be some function of t and X . What is the stochastic differential for $f(t, X)$?
- ▶ If X is a deterministic function of time t , the answer is given by the chain rule of differentiation, that is

$$df(t, X) = \frac{\partial}{\partial t} f dt + \frac{\partial}{\partial X} f dX$$

- ▶ Things are not quite like this if X is a stochastic process

Example

- ▶ Consider a standard Brownian motion W . The stochastic differential is the simplest possible, i.e.

$$dW = 0 \cdot dt + 1 \cdot dW$$

- ▶ Consider the process $Z(t) = f(W(t)) = W^2(t)$. Clearly, this is a positive process.
- ▶ If we apply ordinary calculus rules, we obtain

$$df = 2WdW \quad \text{or in integral form} \quad f(W(t)) = 2 \int_0^t W(s)dW(s)$$

- ▶ Paths of such a process are simulated in "companion 02" script: we see that they can very well become negative.
- ▶ We may conclude that ordinary calculus cannot be applied right away to stochastic processes.

Itô Formula: the result

Assume that the process X has a stochastic differential

$$dX = \mu dt + \sigma dW,$$

where μ, σ are adapted processes.

Let $f : [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ be a $C^{1,2}$ -function and define the process Z by

$$Z(t) = f(t, X(t)).$$

Then Z has a stochastic differential given by

$$\begin{aligned} df(t, X) &= \underbrace{\frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX}_{\text{ordinary calculus}} + \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial x^2} dX dX}_{\text{Itô correction}} \\ &= \left\{ \frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right\} dt + \sigma \frac{\partial f}{\partial x} dW. \end{aligned} \tag{9}$$

Heuristic "proof"

Examples

- ▶ Let's get back to $Z(t) = W(t)^2$. Applying (9) with $f(x) = x^2$, $\mu = 0$ and $\sigma = 1$ we get

$$dW^2 = dt + 2WdW$$

Let's adopt a more compact notation and write X_t instead of $X(t)$.

- Consider now the process

$$X_t = \sigma \int_0^t e^{-as} dW_s$$

where a and σ are real numbers. Consider the process

$$Z_t = f(t, X_t) = e^{at} X_t.$$

Then

$$dZ_t = ae^{at} X_t dt + \sigma dW_t.$$

Substituting $e^{at} X_t = Z_t$ we obtain that Z satisfies the following differential **equation**

$$dZ_t = aZ_t dt + \sigma dW_t. \quad (10)$$

An important example

- Assume that the log-price of an asset $X = \log S$ has differential

$$dX_t = \mu_t dt + \sigma_t dW_t, \quad X_0 = x_0. \quad (11)$$

where μ and σ are adapted and regular processes.

- The price process $S = e^X$ is then given by

$$S_t = S_0 e^{\int_0^t \mu_s ds + \int_0^t \sigma_s dW_s} \quad (12)$$

where $S_0 = e^{x_0}$.

- As for dS , the price process $S = e^X$ has stochastic differential given by

$$dS_t = \left(\mu_t + \frac{1}{2} \sigma_t^2 \right) S_t dt + \sigma_t S_t dW_t.$$

In fact

- ▶ If the processes μ and σ in (11) above are deterministic and constant, then the process S becomes

$$S_t = S_0 e^{\mu t + \sigma W_t} \quad S_0 = e^{x_0}$$

and is termed the *Geometric Brownian motion*.

- ▶ Clearly, it satisfies the differential equation

$$dS = \left(\mu + \frac{1}{2}\sigma^2 \right) S dt + \sigma S dW. \quad (13)$$

- ▶ Let's look at ordinary calculus again

Diffusion Processes and Stochastic Differential Equations

One-dimensional Diffusion

- ▶ A (one-dimensional) **diffusion process** is an Itô process X with stochastic differential of the following form

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t, \quad X(0) = x_0,$$

where W is a Wiener process, while

$$\mu : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R} \quad \text{and} \quad \sigma : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$$

are **sufficiently regular** functions.

- ▶ In other words, the process X satisfies the integral equation

$$X_t = x_0 + \int_0^t \mu(s, X_s)ds + \int_0^t \sigma(s, X_s)dW_s \quad \text{for all } t \geq 0.$$

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t \quad (14)$$

- ▶ The differential equation above is called a **Stochastic Differential Equation** (SDE for short)
- ▶ The local expected change per unit of time μ is termed the **drift** coefficient/term.
- ▶ The local standard deviation per unit of time σ is termed the **diffusion** coefficient/term.

Examples

- ▶ Generalized Brownian Motion

$$dX = \mu dt + \sigma dW \quad X(0) = x_0 \quad \mu, \sigma \in \mathbb{R}$$

with solution

$$X_t = x_0 + \mu t + \sigma W_t.$$

- ▶ Geometric Brownian Motion

$$dX = \mu X dt + \sigma X dW \quad X(0) = x_0 \quad \mu, \sigma \in \mathbb{R}$$

with solution

$$X_t = x_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}.$$

► Linear SDE

$$dX = aXdt + \sigma dW \quad X(0) = x_0 \quad a \in \mathbb{R}$$

with solution

$$X_t = e^{at} x_0 + \sigma e^{at} \int_0^t e^{-as} dW_s$$

Existence and Uniqueness of Solutions

Consider the SDE

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t \quad (15)$$

Theorem (5.1)

Assume there exists a constant K such that the following conditions are satisfied for all x, y and t

$$|\mu(t, x) - \mu(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq K|x - y|, \quad (16)$$

$$|\mu(t, x)| + |\sigma(t, x)| \leq K(1 + |x|). \quad (17)$$

Then there exists a unique solution to (15) such that

- ▶ X is $\mathcal{F}_t^W$ -adapted;
- ▶ X has P -a.s. continuous trajectories;
- ▶ X is a Markov process;
- ▶ X satisfies (5.9) in Björk.

Yet Another Example: the Vasicek Model

- ▶ Consider the following SDE

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t, \quad X_0 = x_0$$

with $\kappa > 0$. Typically $\theta, \sigma > 0$.

- ▶ The solution X of this SDE is an example of a *mean reverting* process:
 - If X at a point in time t is below the level θ , then the drift will push it up;
 - while if it is above, the drift will push the process down to the θ level.

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t, \quad X_0 = x_0$$

- ▶ θ is called the mean-reversion level, while κ is called the speed of mean reversion.
- ▶ This process is often used to describe the short interest rate (the Vasicek model).
- ▶ The solution is given by

$$X_t = e^{-\kappa t} \left(x_0 + \int_0^t \theta e^{\kappa s} ds + \int_0^t \sigma e^{\kappa s} dW_s \right) \quad (18)$$

Euler discretization

- In far the most cases we cannot find the explicit solution of a SDE

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t, \quad X_0 = x_0$$

- However, the SDE gives us the "recipe" on how to approximate the solution by *Euler discretization scheme*.
- To approximate the solution, we consider **discrete** time intervals $\Delta t = t_{i+1} - t_i$, so the equation above becomes

$$X_{t_{i+1}} - X_{t_i} \approx \mu(t_i, X_{t_i})\Delta t + \sigma(t_i, X_{t_i})\Delta W \quad \Delta W \sim \sqrt{\Delta t}\mathcal{N}(0, 1)$$

that is

$$X_{t_{i+1}} \approx X_{t_i} + \mu(t_i, X_{t_i})\Delta t + \sigma(t_i, X_{t_i})\Delta W$$

To generate a path, we may implement the following simulation scheme

```

 $X_0$    =    $x_0$ 
for  $i$    =    $0 : n - 1$ 
 $X_{t_{i+1}}$   $\approx$   $X_{t_i} + \mu(t_i, X_{t_i})\Delta t + \sigma(t_i, X_{t_i})\sqrt{\Delta t}\phi_{i+1},$      $\phi_{i+1} = \mathcal{N}(0, 1)$ 
end
```

How well does Euler approx. work?

To check how well the Euler scheme works perform the following experiment

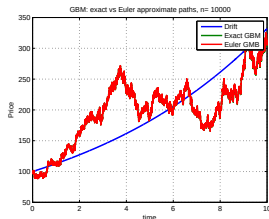
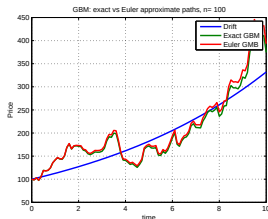
- ▶ Consider a SDE for which the solution is known, e.g. the GBM:

$$dS_t = S_t \mu dt + S_t \sigma dW, \quad S(0) = S_0 \quad (19)$$

$$S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t} \quad (20)$$

- ▶ We have two ways to simulate paths:
 - Exact paths, i.e. generate $X_t = (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$ and then exponentiating as in (20)
 - Via Euler scheme, by discretizing (19)
- ▶ A comparison for different sampling frequency n is carried out in "Companion02"

Exact vs Euler



(see "Companion02")

- ▶ The Euler scheme is very easy to implement.
- ▶ Basically, universally applicable.
- ▶ However, not always sufficiently accurate, but enough for the purpose of this course.
- ▶ Refinements are available, and possible presented in the MC course.

Summing up

- ▶ The vast majority of financial models is constructed upon SDEs

$$dS = \mu(t, S)dt + \sigma(t, S)dW$$

- ▶ For example S could be the price of an asset, the log-price, the short-term interest rate, a volatility process etc.etc.
- ▶ In few cases, it is possible to derive the solution of the SDE, but most often that's not the case.
- ▶ However, the SDE itself contains a lot of information about the process itself and the Euler-scheme is a powerful tool which can be used for a number of purposes - and derivatives pricing as well, as we shall see.
- ▶ **Remark** Quite often we shall meet SDEs "less regular" than those described so far, i.e. with $\mu(t, x)$ and $\sigma(t, x)$ not satisfying the conditions (16) and (17) above. More general existence and uniqueness conditions are of course available in the literature, but such a discussion is outside the scope of this course.

Multidimensional case

Standard d -dimensional Brownian Motion

- ▶ A (standard) d -dimensional Wiener process is a process $\mathbf{W} = (W_1, \dots, W_d)^T$ with the following properties
 - each W_i is a one-dimensional Wiener process;
 - the components W_i are mutually independent.
- ▶ Alternatively, a (standard) d -dim. Wiener process $\mathbf{W}$ is a d -dim. process s.t.
 - $\mathbf{W}_0 = 0$;
 - For $r < s \leq t < u$ we have that $\mathbf{W}_s - \mathbf{W}_r \perp \mathbf{W}_t - \mathbf{W}_u$;
 - d - dimensional standard Normally distributed increments: for all $t > s \geq 0$: $\mathbf{W}_t - \mathbf{W}_s \sim \sqrt{t-s} \mathcal{N}^d(0, \mathbf{I})$, where $\mathbf{I}$ represents the identity matrix.

General d -dimensional Brownian Motion

A d -dimensional Wiener process with correlation matrix $\boldsymbol{\rho} = (\rho_{ij})_{i,j=1,\dots,d}$ is d -dim. process s.t.

- ▶ $W_0 = 0$
- ▶ For $r < s \leq t < u$ we have that $W_s - W_r \perp W_t - W_u$;
- ▶ d - dimensional Normally distributed increments: for all $t > s \geq 0$:
 $W_t - W_s \sim \sqrt{t-s} \mathcal{N}^d(0, \boldsymbol{\rho})$, where the covariance matrix is given by the correlation $\boldsymbol{\rho}$, i.e. $\boldsymbol{\rho}$ is symmetric, positive definite matrix s. t.

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & \rho_{12} & \dots & \rho_{1d} \\ \rho_{12} & 1 & \dots & \rho_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1d} & \rho_{12} & \dots & 1 \end{bmatrix}.$$

with $|\rho_{ij}| < 1$ for the off-diagonal elements.

n -dimensional Itô process

A multidimensional Itô process $\mathbf{X} = (X_1, \dots, X_n)^\top$ is a process with differential

$$d\mathbf{X} = \boldsymbol{\mu}dt + \boldsymbol{\sigma}d\mathbf{W},$$

where

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix}, \boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \dots & \sigma_{1d} \\ \dots & \ddots & \dots \\ \sigma_{n1} & \dots & \sigma_{nd} \end{bmatrix}, \mathbf{W} = \begin{bmatrix} W_1 \\ \vdots \\ W_d \end{bmatrix}.$$

- ▶ $\mathbf{W}$ is a d -dimensional Wiener process with correlation matrix $\boldsymbol{\rho}$
- ▶ $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are vector and matrix processes adapted to $\mathcal{F}^{\mathbf{W}}$.
- ▶ $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are sufficiently integrable .

From general to standard Brownian motions

- Consider a n -dimensional Ito process

$$d\mathbf{X} = \boldsymbol{\mu}dt + \boldsymbol{\sigma}d\mathbf{W}_t,$$

where $\mathbf{W}$ has correlation matrix $\boldsymbol{\rho}$.

- Then we may use the **Cholesky decomposition** of $\boldsymbol{\rho}$. That is, there exists a unique lower triangular matrix $\boldsymbol{\delta}$ such that

$$\boldsymbol{\delta}\boldsymbol{\delta}^T = \boldsymbol{\rho}$$

- We can now express $\mathbf{W}$ as following

$$\mathbf{W} = \boldsymbol{\delta}\widetilde{\mathbf{W}} \tag{21}$$

where $\widetilde{\mathbf{W}}$ is a standard n dimensional Brownian motion.

- Finally

$$d\mathbf{X} = \boldsymbol{\mu}dt + (\boldsymbol{\sigma}\boldsymbol{\delta})d\widetilde{\mathbf{W}}_t,$$

Example

- ▶ Consider a 2-dimensional Brownian Motion $W = (W_1, W_2)$ with correlation matrix

$$\rho = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

- ▶ The Cholesky decomposition of ρ is given by

$$\delta = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{bmatrix}$$

Multidimensional Ito Formula

- ▶ Consider a n -dimensional Ito process

$$d\mathbf{X} = \boldsymbol{\mu}dt + \boldsymbol{\sigma}d\mathbf{W},$$

- ▶ Let $f : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a $C^{1,2}$ function.

- ▶ Then

$$df = \frac{\partial f}{\partial t}dt + \sum_{i=1}^n \frac{\partial f}{\partial x_i}dX_i + \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}dX_i dX_j.$$

with the "operational rules":

$$dtdt = 0, \quad dtd\mathbf{W} = d\mathbf{W}dt = 0, \quad dW_i dW_j = \rho_{i,j}dt \quad (22)$$

Example

- ▶ Consider a 2-dimensional Itô process of the following form

$$dX_1 = \mu_1 dt + \sigma_1 dW_1 \quad (23)$$

$$dX_2 = \mu_2 dt + \sigma_2 dW_2 \quad (24)$$

where μ_1, μ_2, σ_1 and σ_2 are sufficiently regular scalar processes, and $W = (W_1, W_2)$ is a 2-dimensional Brownian Motion with correlation matrix

$$\rho = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

- ▶ Consider the process $Z = X_1 \cdot X_2$.
- ▶ Then Z has differential

$$dZ = X_2 dX_1 + X_1 dX_2 + dX_1 dX_2 \quad (25)$$

$$= X_2 dX_1 + X_1 dX_2 + \rho \sigma_1 \sigma_2 dt \quad (26)$$

and to complete, substitute the expressions for dX_1 and dX_2 .

Multivariate SDEs

Multivariate SDEs

Let $M(n, d)$ denote the class of $n \times d$ matrices, and let

- ▶ W be a d -dimensional Wiener process (column-vector);
- ▶ $\mu : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the drift function;
- ▶ $\sigma : [0, \infty) \times \mathbb{R}^n \rightarrow M(n, d)$;
- ▶ $x_0 \in \mathbb{R}^n$.

A (multivariate) diffusion process X is a process that satisfies the **stochastic differential equation**

$$\begin{cases} dX = \mu(t, X)dt + \sigma(t, X)dW, & t \geq 0, \\ X(0) = x_0, \end{cases} \quad (27)$$

or, equivalently,

$$X_s = x_0 + \int_0^s \mu(s, X_s)ds + \int_0^s \sigma(s, X_s)dW_s, \quad t \geq 0. \quad (28)$$

Existence and uniqueness of solutions to SDEs

Theorem Let measurable functions μ and σ satisfy for $t \geq 0$ Lipschitz and growth conditions

$$\begin{aligned}\|\mu(t, x) - \mu(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\| &\leq L\|x - y\|, \\ \|\mu(t, x)\| + \|\sigma(t, x)\| &\leq L(1 + \|x\|).\end{aligned}$$

for some $L > 0$, where the matrix norm is defined as

$$\|\sigma\| = \sqrt{\sum_{i,j} \sigma_{ij}^2}$$

Then there exists a unique solution to (27) such that

- ▶ X is $\mathcal{F}_t^W$ -adapted;
- ▶ X has $\mathbb{P}$ -a.s. continuous trajectories;
- ▶ X is a Markov process;

Stochastic volatility models

The typical **Stochastic volatility model** takes the following form:

$$dv = a(v)dt + b(v)dW_1 \quad (29)$$

$$dS = \mu Sdt + \sqrt{v}SdW_2 \quad (30)$$

where

- ▶ S represents the price of an asset,
- ▶ v is a process representing the random instantaneous variance,
- ▶ μ is a constant representing the expected return,
- ▶ (W_1, W_2) is a bi-variate Brownian Motions with correlation ρ .

Famous examples include

- ▶ Hull and White (1987):

$$dv = \alpha v dt + \epsilon v dW_1$$

- ▶ Heston (1993)

$$dv = \kappa(\theta - v)dt + \epsilon\sqrt{v_t}dW_1$$

- ▶ Bakshi, Ju, Yang (2006):

$$dv = (\alpha_0 + \alpha_1 v + \alpha_2 v^2 + \alpha_3 v^{-1})dt + \beta^1 v^{\beta^2} dW_1$$

Paths Simulation

- ▶ Applying Itô formula with $X = \log S$, we can re-formulate an SV model in terms of log-price instead to obtain:

$$dv = a(v)dt + b(v)dW_1 \quad (31)$$

$$dX = \left(\mu - \frac{1}{2}v\right)dt + \sqrt{v}dW_2 \quad (32)$$

- ▶ To simulate paths, we can apply a multivariate Euler discretization scheme
- ▶ For a partition of $[0, T]$ in N points, $\Delta t = T/N$, given (v_{t_i}, X_{t_i}) generate the values at the next t_{i+1} as

$$v_{t_{i+1}} = v_{t_i} + a(v_{t_i})\Delta t + b(v_{t_i})\sqrt{\Delta t}\phi_1$$

$$X_{t_{i+1}} = X_{t_i} + \left(\mu - v_{t_i}\right)\Delta t + \sqrt{v_{t_i}}\sqrt{\Delta t}\phi_2$$

where ϕ_1, ϕ_2 are standard normal, with correlation ρ .

- ▶ To generate the correlated ϕ_1, ϕ_2 , proceed via Cholesky-factorization, i.e.
 - Generate two independent standard normal $\varepsilon_1, \varepsilon_2$
 - Set $\phi_1 = \varepsilon_1$ and $\phi_2 = \rho\varepsilon_1 + \sqrt{1 - \rho^2}\varepsilon_2$